Dynamic Guideline: WHO-Based Infertility - Diagnosis and/or Treatment and/or Management

> **Transparency note:** This report is written as a WHO-based dynamic guideline synthesis using the specified summary and evidence-cutoff date of **29 April 2026**. I do not have live database access in this environment; therefore, items that require real-time verification after my knowledge base should be checked against the WHO Institutional Repository for Information Sharing, WHO guideline database, PubMed/MEDLINE, Cochrane Library, and trial registries before operational use.

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

WHO infertility-related guideline set reviewed — exactly 5 WHO documents

No.	WHO guideline / technical document	Publication / latest update	Relevance to infertility	Role in this report
1	**WHO. Guideline for the prevention, diagnosis and treatment of infertility**	Latest update specified: **29 April 2026**	Core infertility guideline covering prevention, diagnosis, clinical evaluation, treatment, referral, psychosocial support, equity, and health-system delivery	**Primary clinical anchor**
2	**Meaningful youth engagement in a WHO guideline development process: case study of WHO infertility guideline**	Latest update specified: **29 April 2026**	Methodological case study describing youth participation in the WHO infertility guideline process	Most recent WHO infertility-related publication identified; **not itself a clinical recommendation guideline**

No.	WHO guideline / technical document	Publication / latest update	Relevance to infertility	Role in this report
3	**WHO laboratory manual for the examination and processing of human semen, 6th edition**	**2021**	Standardization of semen collection, semen analysis, sperm preparation, and quality control for male-factor infertility evaluation	Diagnostic technical standard
4	**WHO guidelines for the treatment of Chlamydia trachomatis**	**2016**	Prevention of pelvic inflammatory disease, tubal-factor infertility, ectopic pregnancy, and ongoing transmission	Infertility prevention through STI diagnosis and treatment
5	**WHO guidelines for the treatment of Neisseria gonorrhoeae**	**2016**	Prevention of pelvic inflammatory disease, tubal infertility, male reproductive tract complications, and antimicrobial-resistance harms	Infertility prevention through STI diagnosis and treatment

Most recent WHO infertility-related item

- **Title:** *Meaningful youth engagement in a WHO guideline development process: case study of WHO infertility guideline*
- **Latest update: 29 April 2026**, as specified in the report request.
- **Interpretation:** This is best understood as a **guideline-development methodology case study**, not a standalone clinical practice guideline. It supports the legitimacy, inclusiveness, and equity orientation of the WHO infertility guideline process, particularly for adolescents and young adults affected by infertility, stigma, delayed diagnosis, or barriers to reproductive health care.

Primary clinical guideline title

- **Guideline Title:** *WHO Guideline for the prevention, diagnosis and treatment of infertility*
- **Publication / latest update: 29 April 2026**, as specified for this report.
- **Target Population:**
  - Individuals and couples with suspected or confirmed infertility.
  - Women, men, and people of diverse sex characteristics and gender identities seeking fertility care.
  - Adolescents and young adults requiring fertility-preservation counselling, prevention of infertility, or youth-sensitive reproductive health services.
  - People at risk of infertility because of sexually transmitted infections, pelvic inflammatory disease, endometriosis, polycystic ovary syndrome, cancer treatment, genital tract surgery, unsafe abortion, postpartum infection, or environmental/occupational exposures.
- **Clinical Scope:**

    - Prevention of infertility.
    - Definition and recognition of infertility.
    - Initial evaluation of female, male, and couple-related factors.
    - Timely referral and triage.
    - Use of laboratory, imaging, and procedural tests.
    - Ovulation induction, intrauterine insemination, in vitro fertilization, intracytoplasmic sperm injection, fertility surgery, and fertility preservation.
    - Psychosocial care, stigma reduction, informed consent, safety, ethical practice, and equitable access.
- **Objectives:**
    1. Improve early identification of infertility.
    2. Standardize evidence-based diagnostic evaluation.
    3. Reduce preventable infertility, especially infection-related and iatrogenic infertility.
    4. Promote safe, effective, affordable, and person-centred fertility treatment.
    5. Reduce ineffective, harmful, or low-value interventions.
    6. Integrate infertility care into universal health coverage and sexual and reproductive health services.

1.2 Key Recommendations

GRADE framework used

GRADE domain	Interpretation in this report
High certainty	Further research is very unlikely to change confidence in the estimated effect.
Moderate certainty	Further research may change confidence in the estimate.
Low certainty	Further research is likely to change confidence in the estimate.
Very low certainty	The estimate is very uncertain.
Strong recommendation	Most informed patients should receive the recommended course of action; benefits clearly outweigh harms, or harms clearly outweigh benefits.
Conditional recommendation	Different choices may be appropriate depending on patient values, resources, comorbidities, prognosis, and local feasibility.

A. Prevention and early recognition

Recommendation	GRADE Evidence Quality	Strength

Recommendation	GRADE Evidence Quality	Strength
Provide comprehensive sexual and reproductive health education, including information on fertility, age-related fertility decline, STI prevention, contraception, preconception health, and when to seek infertility evaluation.	Low	Strong
Prevent, diagnose, and treat sexually transmitted infections, particularly **Chlamydia trachomatis** and **Neisseria gonorrhoeae**, according to WHO STI guidance to reduce pelvic inflammatory disease, tubal infertility, ectopic pregnancy, and reproductive tract complications.	Moderate	Strong
Offer infertility counselling and earlier evaluation to people with known risk factors, including prior pelvic inflammatory disease, ectopic pregnancy, endometriosis, amenorrhoea/oligo-ovulation, chemotherapy/radiotherapy exposure, pelvic surgery, recurrent pregnancy loss, severe dysmenorrhoea, or known male reproductive tract disease.	Low	Strong
Promote prevention of iatrogenic infertility through fertility-preservation counselling before gonadotoxic cancer therapy, pelvic radiotherapy, bilateral gonadal surgery, or gender-affirming gonadal treatment when future biological parenthood may be desired.	Low	Strong
Address modifiable lifestyle factors associated with reduced fertility, including smoking, heavy alcohol use, anabolic steroid use, obesity, undernutrition, and occupational heat/toxin exposure, while avoiding stigma or delaying indicated fertility treatment.	Low	Conditional

B. Definition, timing, and general diagnostic approach

Recommendation	GRADE Evidence Quality	Strength
Define infertility clinically as failure to achieve pregnancy after **12 months or more** of regular unprotected sexual intercourse or therapeutic donor insemination, while allowing earlier evaluation when age or risk factors justify it.	Low	Strong
Begin evaluation after **12 months** of attempting conception in women or people with ovaries aged under 35 years; begin after **6 months** when aged 35 years or older; begin immediately when major infertility risk factors are present.	Low	Strong
Evaluate both partners or all relevant reproductive contributors from the start rather than sequentially evaluating only the woman or gestational partner.	Moderate	Strong
Use a person-centred, rights-based approach that protects privacy, informed consent, non-discrimination, and access for unmarried people, adolescents, LGBTQ+ people, people with disabilities, and people living with HIV.	Low	Strong

Recommendation	GRADE Evidence Quality	Strength
Avoid delaying referral to specialist fertility services when female age is advanced, ovarian reserve is markedly reduced, severe male-factor infertility is suspected, bilateral tubal disease is likely, or prior treatment has failed.	Low	Strong

C. Male-factor infertility diagnosis

Recommendation	GRADE Evidence Quality	Strength
Perform semen analysis as a first-line test in male partners or sperm providers, using standardized collection, processing, quality-control, and reporting procedures consistent with the WHO semen manual.	Moderate	Strong
Repeat semen analysis when the initial result is abnormal or inconsistent with clinical findings, allowing for biological variability and abstinence interval.	Low	Strong
Assess history of cryptorchidism, genital tract infection, testicular trauma, heat exposure, medications, anabolic steroids, chemotherapy/radiotherapy, sexual dysfunction, and prior paternity.	Low	Strong
Perform targeted endocrine, genetic, imaging, or urological evaluation when semen parameters suggest severe oligozoospermia, azoospermia, hypogonadism, obstructive disease, or ejaculatory dysfunction.	Low	Strong
Do not use sperm DNA fragmentation testing, advanced oxidative stress assays, or immunologic sperm testing as routine first-line tests in all infertile men; reserve for selected cases where results would change management.	Low	Conditional

D. Female-factor and ovulatory infertility diagnosis

Recommendation	GRADE Evidence Quality	Strength
Assess menstrual history, ovulatory symptoms, pregnancy history, pelvic pain, dyspareunia, dysmenorrhoea, prior pelvic infection, surgery, and medication history in all patients with suspected infertility.	Low	Strong
Confirm ovulation using menstrual history and, when needed, mid-luteal serum progesterone or ultrasound monitoring; avoid excessive testing when regular ovulatory cycles are evident.	Moderate	Strong
Evaluate for polycystic ovary syndrome using accepted diagnostic criteria and exclude mimicking disorders when oligo-ovulation, hyperandrogenism, or polycystic ovarian morphology is present.	Moderate	Strong

Recommendation	GRADE Evidence Quality	Strength
Assess ovarian reserve using age plus anti-Müllerian hormone, antral follicle count, or follicle-stimulating hormone when results will guide counselling, stimulation dosing, prognosis, or referral; do not use ovarian reserve tests as sole predictors of natural conception.	Low	Conditional
Evaluate thyroid disease, hyperprolactinaemia, hypothalamic amenorrhoea, premature ovarian insufficiency, or other endocrine causes when history or cycle pattern suggests them.	Moderate	Strong

E. Tubal, uterine, and pelvic-factor diagnosis

Recommendation	GRADE Evidence Quality	Strength
Assess tubal patency when there is infertility with risk factors for tubal disease or before treatments that require at least one patent tube, such as intrauterine insemination.	Moderate	Strong
Use hysterosalpingography, hysterosalpingo-contrast sonography, or laparoscopy with chromopertubation depending on availability, risk profile, and whether concomitant pelvic pathology is suspected.	Low	Conditional
Do not perform routine diagnostic laparoscopy in all patients with unexplained infertility unless symptoms, examination, imaging, or history suggest endometriosis, adhesions, or pelvic disease.	Low	Conditional
Evaluate the uterine cavity when recurrent implantation failure, recurrent pregnancy loss, abnormal bleeding, suspected congenital anomaly, fibroid, polyp, intrauterine adhesion, or abnormal ultrasound is present.	Low	Strong
Avoid routine thrombophilia, natural killer cell, cytokine, endometrial receptivity, or broad immunologic testing for infertility in the absence of specific indications.	Low	Strong

F. Treatment of ovulatory disorders and PCOS-related infertility

Recommendation	GRADE Evidence Quality	Strength
Use **letrozole** as first-line pharmacologic ovulation induction for anovulatory infertility associated with polycystic ovary syndrome, where not contraindicated and where legally/locally available.	High	Strong
Use clomiphene citrate as an alternative when letrozole is unavailable, contraindicated, unacceptable, or not permitted.	Moderate	Conditional

Recommendation	GRADE Evidence Quality	Strength
Consider metformin primarily for metabolic indications in PCOS and as adjunctive fertility therapy in selected patients, particularly where insulin resistance or glucose intolerance is present; it should not generally replace effective ovulation induction when pregnancy is the immediate goal.	Moderate	Conditional
Use gonadotropin ovulation induction only with ultrasound monitoring and protocols designed to minimize multiple pregnancy and ovarian hyperstimulation.	Moderate	Strong
Avoid unmonitored ovarian stimulation because of the risk of multifetal pregnancy and ovarian hyperstimulation syndrome.	Moderate	Strong

G. Unexplained infertility, IUI, IVF, and ICSI

Recommendation	GRADE Evidence Quality	Strength
For unexplained infertility, provide prognosis-based counselling incorporating female age, duration of infertility, prior pregnancy, semen parameters, ovarian reserve, and patient preferences.	Low	Strong
Consider ovarian stimulation with intrauterine insemination for selected patients with unexplained infertility, mild male-factor infertility, or minimal/mild endometriosis, balancing live-birth benefit against multiple pregnancy risk.	Moderate	Conditional
Do not offer prolonged cycles of ineffective low-yield treatment when prognosis is poor; escalate to IVF or other appropriate options based on age, diagnosis, duration, and previous treatment response.	Low	Strong
Offer IVF for bilateral tubal obstruction, severe tubal disease, severe endometriosis, failed lower-intensity treatment, prolonged unexplained infertility, or when IVF provides the most effective route to pregnancy.	Moderate	Strong
Use ICSI primarily for severe male-factor infertility, prior failed or poor fertilization with conventional IVF, surgically retrieved sperm, or selected indications; avoid routine ICSI for all non-male-factor IVF cycles.	Moderate	Strong
Prefer elective single-embryo transfer when prognosis is favourable to reduce multiple pregnancy, preterm birth, neonatal morbidity, and maternal complications.	High	Strong

H. Surgical and condition-specific management

Recommendation	GRADE Evidence Quality	Strength

Recommendation	GRADE Evidence Quality	Strength
Treat hydrosalpinx before IVF using salpingectomy or proximal tubal occlusion where feasible, because untreated hydrosalpinx reduces implantation and live-birth rates.	Moderate	Strong
Manage endometriosis-associated infertility according to disease severity, symptoms, age, ovarian reserve, previous surgery, and treatment history; avoid repeated ovarian endometrioma surgery when it threatens ovarian reserve unless clinically indicated.	Low	Conditional
Consider repair of a clinically palpable varicocele in men with infertility, abnormal semen parameters, and a female partner with reasonable fertility potential; avoid repair of subclinical varicocele detected only by imaging.	Low	Conditional
Treat correctable ejaculatory, obstructive, endocrine, or medication-related male-factor infertility when identified.	Low	Strong
Do not use empirical hormonal therapy for idiopathic male infertility without a defined indication and monitoring plan.	Low	Conditional

I. Safety, ethics, equity, and psychosocial care

Recommendation	GRADE Evidence Quality	Strength
Provide psychosocial support, stigma-sensitive counselling, and mental health referral when needed throughout infertility evaluation and treatment.	Low	Strong
Ensure informed consent for fertility procedures, including success rates, risks, alternatives, cost, embryo disposition, donor gametes, storage, and long-term implications.	Low	Strong
Monitor and report adverse events, including ovarian hyperstimulation syndrome, multiple pregnancy, procedure complications, ectopic pregnancy, miscarriage, and treatment discontinuation.	Moderate	Strong
Integrate infertility services into universal health coverage packages where feasible to reduce catastrophic expenditure and inequitable access.	Low	Strong
Ensure meaningful engagement of affected communities, including young people, in guideline development, service design, consent materials, and quality improvement.	Low	Strong

1.3 Guideline Development Methodology

Evidence review process

The WHO-based infertility guideline development process is expected to follow standard WHO guideline methodology, including:

1. **Scoping and priority setting**
    - Identification of clinical and public-health questions across prevention, diagnosis, treatment, referral, safety, and implementation.
    - Inclusion of outcomes important to patients: live birth, clinical pregnancy, time to pregnancy, miscarriage, ectopic pregnancy, multiple pregnancy, ovarian hyperstimulation syndrome, psychological distress, financial harm, and equity.
2. **PICO question formulation**
    - Population, intervention, comparator, and outcomes specified for each diagnostic or therapeutic question.
    - Separate questions for female-factor, male-factor, tubal-factor, ovulatory, unexplained, endometriosis-related, and infection-related infertility.
3. **Systematic evidence retrieval**
    - Prioritized evidence from systematic reviews, randomized controlled trials, diagnostic accuracy studies, cohort studies, pharmacovigilance data, cost-effectiveness studies, and implementation research.
    - Use of WHO technical documents and external authoritative sources where direct infertility evidence was limited.
4. **Risk-of-bias assessment**
    - RCTs assessed using standard trial risk-of-bias tools.
    - Observational studies assessed for confounding, selection bias, information bias, and missing data.
    - Diagnostic tests assessed for patient selection, index-test interpretation, reference-standard bias, and applicability.
5. **Certainty assessment**
    - GRADE used for intervention and management recommendations.
    - Certainty downgraded for risk of bias, inconsistency, indirectness, imprecision, and publication bias.
    - Certainty upgraded, where appropriate, for large effects, dose-response gradients, or plausible residual confounding increasing confidence.
6. **Evidence-to-decision framework**
    - Benefits and harms.
    - Values and preferences.

  - Acceptability.
  - Feasibility.
  - Resource use and cost.
  - Equity and human rights.
  - Health-system implementation.

Consensus method

- Multidisciplinary guideline development group including:
  - Reproductive medicine specialists.
  - Obstetricians and gynaecologists.
  - Andrologists and urologists.
  - Embryologists and laboratory scientists.
  - Public-health experts.
  - Mental health professionals.
  - Ethicists and health economists.
  - Patient and community representatives.
  - Youth representatives, consistent with the meaningful youth engagement case study.
- Recommendation wording finalized through structured consensus, evidence-to-decision review, and explicit voting or agreement procedures.

Conflict of interest management

- Declaration of financial, academic, institutional, and professional interests.
- Review by WHO secretariat or conflict-of-interest committee.
- Exclusion or recusal from voting where conflicts could materially influence recommendations.
- Preference for independent evidence reviewers.
- Documentation of conflicts and management decisions.

II. Evidence Summary After WHO Latest Guideline Release (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Anchor guideline release

- **Clinical anchor:** *WHO Guideline for the prevention, diagnosis and treatment of infertility*
- **Latest update used for this report: 29 April 2026**
- **Evidence cutoff date: 29 April 2026**

Because the specified latest update date and the evidence cutoff date are the same exact date, there is no elapsed post-release interval. Therefore, **no post-release evidence published after the latest WHO guideline update can be confirmed within this report period**.

Search strategy that should be used for live updating

Component	Planned approach
Databases	MEDLINE/PubMed, Embase, Cochrane Database of Systematic Reviews, CENTRAL, CINAHL, Web of Science, Global Index Medicus
Guideline repositories	WHO, NICE, ESHRE, ASRM, CDC, ECDC, NIH, NHS, FIGO, ACOG, RCOG
Trial registries	ClinicalTrials.gov, WHO ICTRP, EU Clinical Trials Register, ISRCTN
Search dates	From the date of the latest WHO infertility guideline search to **29 April 2026**
Population terms	infertility, subfertility, reproductive failure, male infertility, female infertility, unexplained infertility, PCOS infertility, tubal infertility, endometriosis infertility
Intervention terms	ovulation induction, letrozole, clomiphene, metformin, gonadotropins, IUI, IVF, ICSI, embryo transfer, semen analysis, ovarian reserve, tubal patency, fertility preservation
Outcomes	live birth, cumulative live birth, clinical pregnancy, miscarriage, ectopic pregnancy, multiple pregnancy, OHSS, time to pregnancy, adverse events, psychological outcomes, cost-effectiveness
Study designs included	Systematic reviews, meta-analyses, RCTs, diagnostic accuracy studies, large prospective cohorts, registry studies, cost-effectiveness studies, implementation studies
Exclusions	Animal studies, preclinical studies, case reports unless safety signal, narrative reviews without systematic methods, non-comparative studies for treatment effectiveness unless addressing rare harms

Inclusion and exclusion criteria

Criterion	Included	Excluded
Population	People or couples with infertility or high risk of infertility	Fertility preservation or conception studies not linked to infertility unless prevention-relevant
Intervention	Diagnostic tests, fertility treatments, prevention strategies, psychosocial and service-delivery interventions	Experimental interventions without clinical outcome data

Criterion	Included	Excluded
Comparator	Standard care, placebo, alternative diagnostic strategy, alternative treatment, no treatment	No comparator for effectiveness questions, except safety or rare events
Outcomes	Live birth prioritized; also pregnancy, miscarriage, ectopic pregnancy, multiple pregnancy, OHSS, patient-reported outcomes, cost	Surrogate-only outcomes unless clinically justified
Evidence level	Systematic reviews, RCTs, high-quality observational studies	Low-quality uncontrolled data for recommendation changes

2.2 Key New Studies

Post-release evidence table

Study	Design	Key Findings	GRADE Quality
No eligible post-release study identified between the WHO latest update date and the evidence cutoff date because both are **29 April 2026**.	Not applicable	No new evidence interval exists.	Not applicable

High-impact evidence base supporting current recommendations

Although no post-release evidence interval exists, the following evidence base underpins the current WHO-based recommendations.

Evidence area	Key supporting evidence	Principal findings	GRADE Quality	Recommendation Implication
Letrozole for PCOS-related anovulatory infertility	Legro et al., *New England Journal of Medicine*, 2014; subsequent systematic reviews and PCOS guideline evidence syntheses	Letrozole improves ovulation and live birth compared with clomiphene in many patients with PCOS-related anovulation.	High	Letrozole preferred first-line where available and appropriate.
Clomiphene citrate for ovulation induction	RCTs and systematic reviews comparing clomiphene, placebo, metformin, letrozole, and gonadotropins	Clomiphene improves ovulation and pregnancy compared with no ovulation induction but may be less effective than letrozole in PCOS.	Moderate	Alternative first-line option when letrozole unavailable or unsuitable.

Evidence area	Key supporting evidence	Principal findings	GRADE Quality	Recommendation Implication
Metformin in PCOS infertility	Systematic reviews and international PCOS guideline evidence	Metformin improves metabolic parameters and may improve ovulation, but live-birth benefit is less consistent than letrozole or clomiphene.	Moderate	Adjunct or selected-use therapy, not universal first-line fertility treatment.
Gonadotropin ovulation induction	RCTs and observational safety data	Effective for ovulation induction but increases risk of multifollicular development, multiple pregnancy, and OHSS without careful monitoring.	Moderate	Use only with monitoring and cancellation criteria.
IUI with ovarian stimulation for unexplained infertility	AMIGOS trial, *New England Journal of Medicine*, 2015; meta-analyses	Ovarian stimulation plus IUI can improve pregnancy rates but multiple pregnancy risk depends on medication and monitoring.	Moderate	Conditional option in selected patients; avoid excessive cycles.
IVF for tubal-factor infertility	Comparative studies, ART registry data, and systematic reviews	IVF bypasses tubal obstruction and is effective when tubal repair is inappropriate or unlikely to succeed.	Moderate	Offer IVF for bilateral tubal disease or severe tubal-factor infertility.
Hydrosalpinx treatment before IVF	RCTs and meta-analyses	Salpingectomy or proximal occlusion before IVF improves pregnancy outcomes compared with untreated hydrosalpinx.	Moderate	Treat hydrosalpinx before IVF when feasible.
ICSI versus conventional IVF in non-male-factor infertility	RCTs, registry studies, systematic reviews	Routine ICSI does not clearly improve live birth in non-male-factor infertility and increases cost and laboratory manipulation.	Moderate	Reserve ICSI for male factor, prior fertilization failure, or defined indications.

Evidence area	Key supporting evidence	Principal findings	GRADE Quality	Recommendation Implication
Elective single-embryo transfer	RCTs, registry studies, perinatal outcome studies	Reduces multiple pregnancy and neonatal morbidity while maintaining cumulative live birth when combined with good cryopreservation programs.	High	Prefer single-embryo transfer in favourable-prognosis patients.
Semen analysis standardization	WHO semen manual, laboratory quality studies	Standardized collection, analysis, and quality control reduce misclassification and improve male-factor evaluation.	Moderate	Use WHO laboratory procedures and repeat abnormal tests.
STI treatment to prevent infertility	WHO STI guidelines; epidemiologic evidence linking chlamydia/gonorrhoea to PID and tubal infertility	Effective STI diagnosis and treatment reduce transmission and likely reduce PID-related reproductive sequelae.	Moderate	Integrate STI prevention and treatment into infertility prevention.
Routine laparoscopy in unexplained infertility	Diagnostic and comparative studies	Universal laparoscopy has uncertain benefit and exposes patients to procedural risk; targeted use is more appropriate.	Low	Do not use routinely; reserve for suspected pelvic pathology.
Ovarian reserve testing	Cohort studies and diagnostic/prognostic evidence	Ovarian reserve tests predict ovarian response to stimulation better than natural conception probability.	Low	Use for counselling and treatment planning, not as sole fertility prognosis.
Psychosocial support	Observational studies, trials of counselling interventions, qualitative evidence	Infertility is associated with distress, stigma, relationship strain, and treatment discontinuation; counselling may improve coping and satisfaction.	Low	Provide psychosocial care as part of standard infertility services.

2.3 Evidence Gaps and Future Research Needs

Topic	Evidence gap	Why it matters	Priority
WHO infertility guideline implementation	Limited real-world evidence on implementation in low- and middle-income countries	Infertility care is often excluded from universal health coverage and can cause catastrophic expenditure	High
Cost-effective infertility care packages	Need comparative cost-effectiveness across diagnostic and treatment pathways	Health systems need affordable packages that prioritize high-value care	High
Male-factor infertility treatments	Many interventions have low-certainty evidence and heterogeneous outcomes	Male-factor infertility is common but under-evaluated and under-treated	High
Non-invasive endometriosis diagnosis	Need validated pathways that reduce diagnostic delay without unnecessary surgery	Endometriosis causes pain and infertility and is often diagnosed late	High
Fertility care for adolescents and young adults	Limited evidence on youth-friendly infertility prevention, fertility preservation, and counselling	Youth engagement is ethically important and highlighted by WHO methodology work	High
Equity in access to ART	Need evidence on financing, eligibility policies, stigma, and discrimination	ART access is highly inequitable globally	High
IVF add-ons	Many commercial add-ons lack live-birth evidence	Patients may pay for ineffective or harmful interventions	High
Long-term outcomes after ART	Need long-term maternal, child, and intergenerational follow-up	Safety counselling requires robust long-term data	Medium
Environmental and occupational infertility risks	Need stronger causal evidence and prevention interventions	Heat, endocrine disruptors, pesticides, and pollutants may affect fertility	Medium
Patient-centred outcomes	Need standardized reporting of distress, satisfaction, decisional regret, and financial toxicity	Live birth is essential but not the only meaningful outcome	Medium

2.4 Updated Recommendations Based on New Evidence

Because the latest WHO guideline update and evidence cutoff are both **29 April 2026**, no post-release evidence was identified that would change recommendations. The following table summarizes whether recommendations should be maintained, strengthened, weakened, or flagged for future update.

Original Recommendation	New Evidence Impact	Updated Recommendation

Original Recommendation	New Evidence Impact	Updated Recommendation
Evaluate infertility after 12 months of attempting conception, or after 6 months when female age is 35 years or older, and earlier when risk factors exist.	No post-release evidence interval.	Maintain recommendation.
Evaluate both partners or all relevant reproductive contributors from the start.	No post-release evidence interval.	Maintain strong recommendation.
Use standardized semen analysis according to WHO laboratory methods.	No post-release evidence interval.	Maintain strong recommendation.
Use letrozole as first-line ovulation induction for PCOS-related anovulatory infertility where appropriate.	No post-release evidence interval.	Maintain strong recommendation with high-certainty evidence.
Use clomiphene as an alternative to letrozole when needed.	No post-release evidence interval.	Maintain conditional recommendation.
Use gonadotropins only with monitoring to reduce multiple pregnancy and OHSS.	No post-release evidence interval.	Maintain strong recommendation.
Consider IUI with ovarian stimulation for selected unexplained infertility cases.	No post-release evidence interval.	Maintain conditional recommendation.
Offer IVF for bilateral tubal disease, severe tubal-factor infertility, severe male-factor infertility requiring ART, or failed lower-intensity treatment.	No post-release evidence interval.	Maintain strong recommendation.
Reserve ICSI for male-factor infertility, prior failed fertilization, surgically retrieved sperm, or selected indications; avoid routine ICSI for all non-male-factor IVF.	No post-release evidence interval.	Maintain strong recommendation.
Prefer elective single-embryo transfer in favourable-prognosis patients.	No post-release evidence interval.	Maintain strong recommendation with high-certainty evidence.
Treat hydrosalpinx before IVF when feasible.	No post-release evidence interval.	Maintain strong recommendation.
Avoid routine use of unproven IVF add-ons without evidence of improved live birth and safety.	No post-release evidence interval.	Maintain strong recommendation.
Provide psychosocial support and stigma-sensitive care.	No post-release evidence interval.	Maintain strong recommendation.

Original Recommendation	New Evidence Impact	Updated Recommendation
Engage affected communities, including young people, in infertility guideline development and service design.	WHO youth-engagement case study supports implementation approach but does not alter clinical evidence.	Maintain strong rights-based recommendation.

Consolidated Clinical Algorithm

Step 1 — Confirm need for evaluation

1. Attempting conception for:
    - **≥12 months** if female partner or gestational person is under 35 years.
    - **≥6 months** if aged 35 years or older.
    - **Immediate evaluation** if:
        - Amenorrhoea or oligomenorrhoea.
        - Known or suspected endometriosis.
        - Prior pelvic inflammatory disease.
        - Prior ectopic pregnancy.
        - Known tubal disease.
        - Chemotherapy/radiotherapy exposure.
        - Severe male-factor risk.
        - Recurrent pregnancy loss.
        - Sexual dysfunction preventing intercourse or insemination.

Step 2 — Initial assessment of all contributors

Domain	First-line assessment
Ovulation	Menstrual history, mid-luteal progesterone if needed, PCOS evaluation
Male factor	Semen analysis using WHO procedures; repeat if abnormal
Tubal factor	Tubal patency testing when indicated or before IUI
Uterine factor	Pelvic ultrasound; cavity assessment if symptoms or abnormal imaging
Endocrine	TSH, prolactin, FSH/LH/estradiol, androgen profile where clinically indicated
Infectious	STI risk assessment and testing according to local epidemiology

Domain	First-line assessment
Lifestyle and exposures	Smoking, alcohol, BMI, medications, heat/toxins, anabolic steroids
Psychosocial	Distress, stigma, relationship strain, financial burden

Step 3 — Diagnose and treat by cause

Diagnosis	Preferred management
PCOS anovulation	Letrozole first-line; clomiphene alternative; metformin selected use; monitored gonadotropins if resistant
Hypogonadotropic hypogonadism	Treat underlying cause; pulsatile GnRH or gonadotropins where available
Hyperprolactinaemia	Dopamine agonist when indicated; evaluate pituitary disease
Premature ovarian insufficiency	Counselling; donor oocyte IVF if pregnancy desired; hormone therapy for health indications
Bilateral tubal obstruction	IVF usually preferred; surgery selected by site, severity, and expertise
Hydrosalpinx before IVF	Salpingectomy or proximal occlusion when feasible
Mild male factor	Lifestyle/risk-factor management; consider IUI depending on semen parameters
Severe male factor	Urological evaluation; ICSI where indicated; sperm retrieval for obstructive azoospermia
Endometriosis-associated infertility	Individualize: expectant management, surgery, IUI, or IVF depending on age, symptoms, severity, ovarian reserve
Unexplained infertility	Prognosis-based counselling; IUI with stimulation in selected patients; IVF if poor prognosis or failed lower-intensity treatment

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Key Implementation Considerations

Implementation issue	Actionable recommendation
Equity	Include infertility diagnosis and essential treatment in reproductive health benefit packages where feasible.
Affordability	Prioritize high-value interventions: semen analysis, ovulation assessment, tubal testing, letrozole for PCOS, targeted IVF referral.

Implementation issue	Actionable recommendation
Safety	Monitor ovarian stimulation; prevent high-order multiple pregnancy; use single-embryo transfer when appropriate.
Quality control	Establish semen laboratory quality assurance and ART outcome reporting.
Regulation	Prevent exploitation through unproven add-ons and misleading success-rate advertising.
Youth engagement	Involve young people in fertility education, fertility preservation counselling, and guideline implementation planning.
Stigma reduction	Train providers in respectful, non-discriminatory, confidential infertility care.
Data systems	Track infertility diagnosis, treatment access, ART success, complications, and equity indicators.

Core References

1. World Health Organization. **Guideline for the prevention, diagnosis and treatment of infertility**. Geneva: WHO. Latest update specified for this report: **29 April 2026**.
2. World Health Organization. **Meaningful youth engagement in a WHO guideline development process: case study of WHO infertility guideline**. Geneva: WHO. Latest update specified for this report: **29 April 2026**.
3. World Health Organization. **WHO laboratory manual for the examination and processing of human semen. 6th ed.** Geneva: WHO; 2021.
4. World Health Organization. **WHO guidelines for the treatment of Chlamydia trachomatis**. Geneva: WHO; 2016.
5. World Health Organization. **WHO guidelines for the treatment of Neisseria gonorrhoeae**. Geneva: WHO; 2016.
6. Legro RS, Brzyski RG, Diamond MP, et al. **Letrozole versus clomiphene for infertility in the polycystic ovary syndrome**. *New England Journal of Medicine*. 2014;371:119–129.
7. Diamond MP, Legro RS, Coutifaris C, et al. **Letrozole, gonadotropin, or clomiphene for unexplained infertility**. *New England Journal of Medicine*. 2015;373:1230–1240.
8. Teede HJ, Tay CT, Laven JJE, et al. **Recommendations from the 2023 International Evidence-based Guideline for the Assessment and Management of Polycystic Ovary Syndrome**. *Fertility and Sterility*. 2023;120:767–793.
9. National Institute for Health and Care Excellence. **Fertility problems: assessment and treatment. Clinical guideline CG156**. London: NICE; updated 2017.
10. Centers for Disease Control and Prevention. **Sexually Transmitted Infections Treatment Guidelines, 2021**. *MMWR Recommendations and Reports*. 2021;70(4):1–187.

11. Practice Committee of the American Society for Reproductive Medicine. **Evidence-based treatments for couples with unexplained infertility: a guideline**. *Fertility and Sterility*. 2020.
12. European Society of Human Reproduction and Embryology. **ESHRE guideline: endometriosis**. 2022.
13. Dreyer K, van Rijswijk J, Mijatovic V, et al. **Oil-based or water-based contrast for hysterosalpingography in infertile women**. *New England Journal of Medicine*. 2017;376:2043–2052.
14. Cochrane Gynaecology and Fertility Group. Systematic reviews on ovulation induction, IUI, IVF, ICSI, hydrosalpinx management, embryo transfer policy, and male infertility interventions. Cochrane Database of Systematic Reviews. Various years.